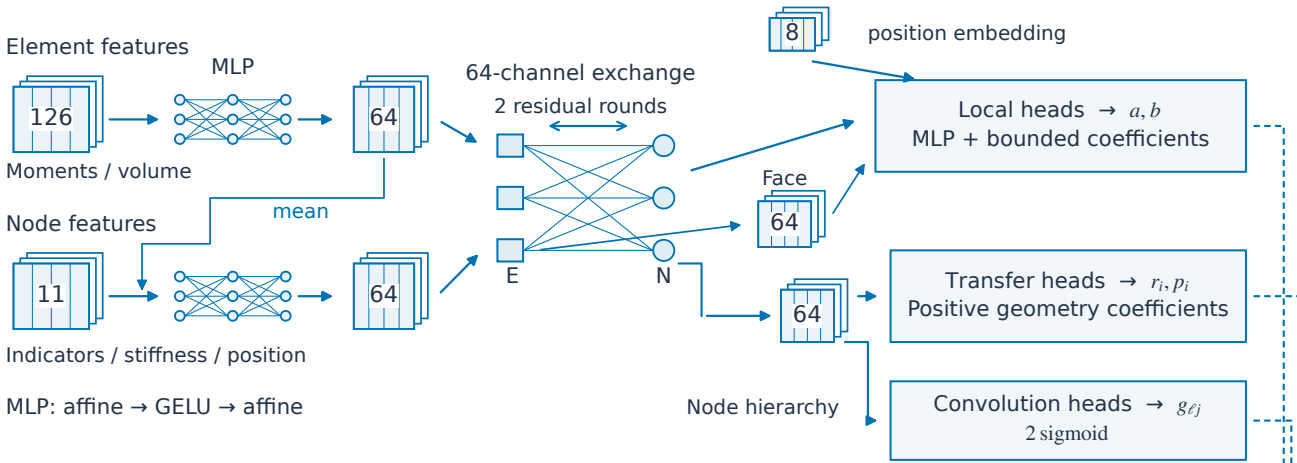
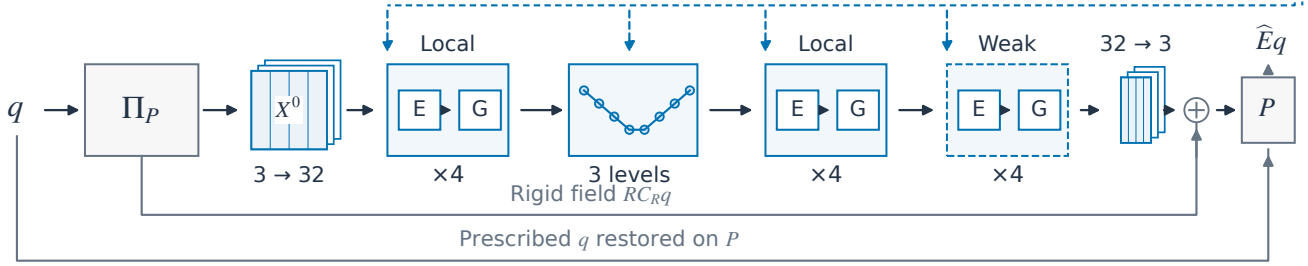


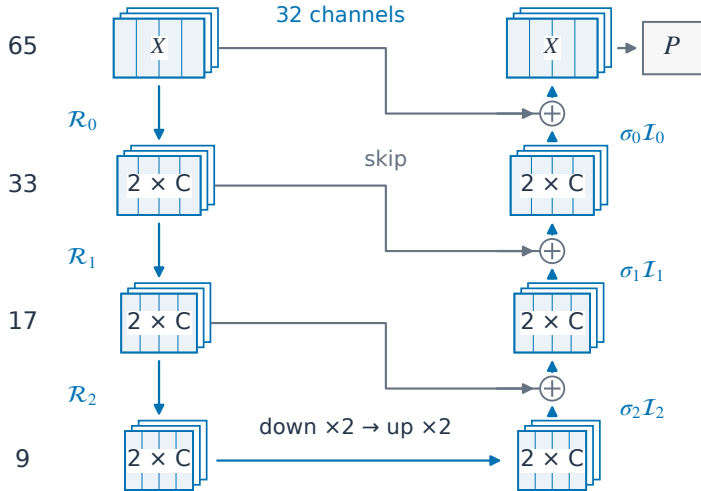
## (a) Geometry branch



## (b) Displacement branch: linear in $q$ for fixed geometry



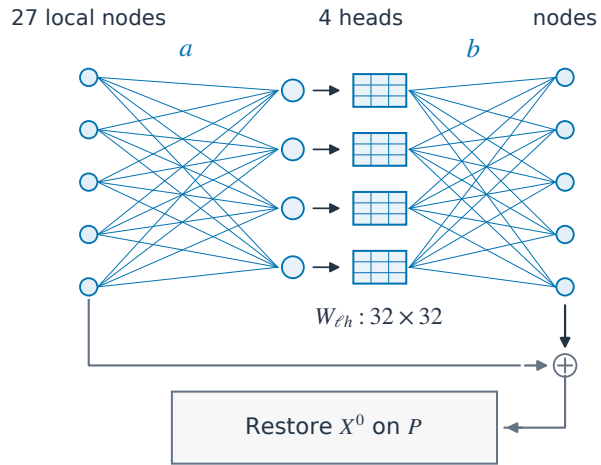
## (c) Multiscale displacement propagation



$\mathcal{R}$ : restriction  $\mathcal{I}$ : prolongation

$$C: X \leftarrow X + g \odot \text{Conv}_{3 \times 3 \times 3}(X)$$

## (d) Local interaction



E: 27 Q2 nodes

G: 18 owner + 9 neighbour nodes

Training: update network parameters  $\theta$  through both branches. Deployment: fix  $\theta$ ; cache geometry coefficients.